



CIS 635 Knowledge Discovery & Data Mining

Vector Space, Distance Metrics, KNN



Outline

- Concept of Vectors and Vector operations
- Proximity vs Distance Metric
- k-NN, our first ML model



Basic Math - Concept of Vectors, and Vector Operations



Basic Math - Concept of Vectors

We are aware of Scalars: A person's

Height (1.72m)



Basic Math - Concept of Vectors

We are aware of Scalars: A person's

Height (1.72m)

Weight (72kg)



Basic Math - Concept of Vectors

We are aware of Scalars: A person's

Height (1.72m)

Weight (72kg)

Salary (100K)



Basic Math - Concept of Vectors

We are aware of Scalars: A person's

Height (1.72m)

Weight (72kg)

Salary (100K)

....



Basic Math - Concept of Vectors

A closed form definition of a person through some features

[Height (1.72m), Weight (72kg), Salary (100K)]



Basic Math - Concept of Vectors

A closed form definition of a person through some features

- no explicit unit mentions

[1.72, 72, 100]



Basic Math - Concept of Vectors

A closed form definition of a person through some features

- no explicit unit mentions

[1.72, 72, 100]

*Is a vectoried representation of a person through
some attributes: height, weight, salary*



Basic Math - Concept of Vectors

A closed form definition of a person through some features

- no explicit unit mentions

[1.72, 72, 100],
[1.65, 70, 120],
[1.81, 110, 90],
...
[1.45, 65, 130],

And here we are talking about *a number of people through same features: height, weight, salary*



Formal definition of Vectors

Basic Math - Concept of Vectors

Formal definition of Vectors

1. Vectors

We begin by defining a mathematical abstraction known as a **vector space**. In linear algebra the fundamental concepts relate to the **n -tuples** and their algebraic properties.

Definition: An ordered n -tuple is considered as a sequence of n **terms** $(a_1, a_2, \dots, a_n)$, where n is a positive integer.

We see that an ordered **n -tuple** has **terms** whereas a set has **members**.

Example: A sequence (5) is called an ordered **1-tuple**. A **2-tuple**, for example (3, 6) (where 6 appears after 3) is called an ordered pair, and **3-tuple** is called an ordered triple. A sequence (9, 3, 4, 4, 1) is called an ordered **5-tuple**.

Let us denote the set of all ordered **1-tuples** of real numbers by $\mathbb{R}$. We will write for example $(3.5) \in \mathbb{R}$.

$$\mathbf{X} = [1.78, 72, 100]$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$



Basic Math - Concept of Vectors

We are aware of Scalars: A person's height, weight, salary

Physics vector: velocity (scalar value + direction)

Algebraic vector (in general): Common representation of an entity (1 to n dimension):

- A person's (height, weight, salary), say [\[1.78, 72, 100\]](#): once defined, we have to follow it.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$



Basic Math - Vector Operations

Vector operation rules

1. $\mathbf{x} + \mathbf{y} \in \mathbb{R}^n$
2. $\alpha \cdot \mathbf{x} \in \mathbb{R}^n$
3. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x} \in \mathbb{R}^n$ *(commutativity)*
4. $\alpha \cdot (\mathbf{x} + \mathbf{y}) = \alpha \cdot \mathbf{x} + \alpha \cdot \mathbf{y}$ *(distributivity)*
5. $(\alpha + \beta) \cdot \mathbf{x} = \alpha \cdot \mathbf{x} + \beta \cdot \mathbf{x}$ *(distributivity)*
6. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$ *(associativity)*
7. $(\alpha\beta) \cdot \mathbf{x} = \alpha \cdot (\beta \cdot \mathbf{x})$ *(associativity)*



Basic Math - Vector Operations

Vector Operation

1.1.2. Vector Addition

Addition of vectors is defined:

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Example:

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 2 \\ 6 \\ -5 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \\ -1 \end{bmatrix}$$



Basic Math - Vector Operations

Vector Operation

1.1.4. Zero Vector

The **zero** vector **sometimes denoted** **0** is a vector having all elements equal to zero, e.g., the 2-dimensional **0** vector:

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (\text{A.7})$$



Basic Math - Vector Operations

Vector Operation

1.1.9. Inner Product

The **inner** or **dot** product of two vectors **x** and **y** of the same dimension is a **scalar** defined by:

$$\mathbf{x}^T \cdot \mathbf{y} = (\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + \cdots + x_ny_n = \sum_{i=1}^n x_iy_i \quad (\text{A.11})$$

Note that the inner product of vector **x** and **y** requires that a transposed vector **x** be multiplied by the **y** vector. Sometimes the inner product is denoted simply by juxtaposition of the vectors **x** and **y**, for example, as $\langle \mathbf{x}, \mathbf{y} \rangle$ or $(\mathbf{x}, \mathbf{y})$.

Example: The inner product of two vectors $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 7 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 0 \\ 2 \\ -3 \end{bmatrix}$

$$\mathbf{x}^T \mathbf{y} = [4 \ 1 \ 7]^T \begin{bmatrix} 0 \\ 2 \\ -3 \end{bmatrix} = 4 \cdot 0 + 1 \cdot 2 + 7 \cdot (-3) = 19$$



Basic Math - Vector Operations

Vector Operation

1.1.10. Orthogonal Vectors

Two vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be **orthogonal** if their inner product is equal to zero

$$\mathbf{x}^T \mathbf{y} = 0 \quad (\text{A.12})$$

here 0 is a scalar.

Example: Two vectors $\mathbf{x} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ and are orthogonal, since their inner product is equal to zero

$$\mathbf{x}^T \cdot \mathbf{y} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}^T = [0 \ 2] = 4 \cdot 0 + 0 \cdot 2 = 0$$



Basic Math - Vector Operations

Vector Operation

1.1.11. Vector Norm

The magnitude of a vector may be measured in different ways. One method, called the vector **norm**, is a function from $\mathbb{R}^n$ into $\mathbb{R}$ for $\mathbf{x}$ an element of $\mathbb{R}^n$. It is denoted $\|\mathbf{x}\|$ and satisfies the following conditions:

1. $\|\mathbf{x}\| \geq 0$, and the equality holds if and only if $\mathbf{x} = \mathbf{0}$
2. $\|\alpha\mathbf{x}\| = |\alpha| \cdot \|\mathbf{x}\|$, where $|\alpha|$ is the absolute value of scalar α

and is defined as:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \quad (\text{A.13})$$

Example: For the vector $\mathbf{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ the norm is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{4^2 + 3^2} = 5$$



Break!



Distance (or Proximity) metric

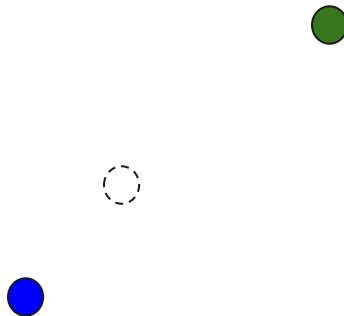
- Let's we are give two data points: one, the **blue**, and the other is the **green** circle.





Distance (or Proximity) metric

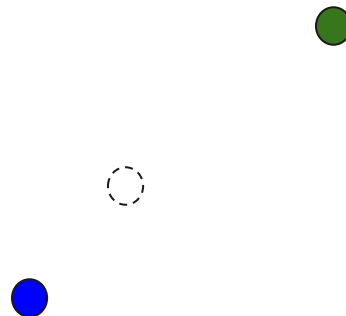
- Let's we are give two data points: one, the **blue**, and the other is the **green** circle.
- Now lets, we are given a new circle, and we are given the task to label it with either blue or green.





Distance (or Proximity) metric

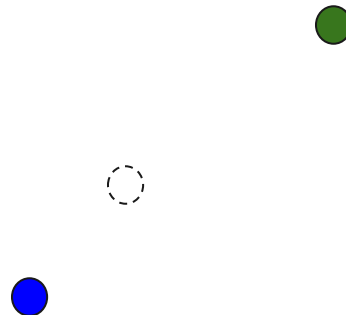
- Let's we are give two data points: one, the **blue**, and the other is the **green** circle.
- Now lets, we are given a new circle, and we are given the task to label it with either blue or green.
- Visually, it looks to be closer to the blue, dot; but how do we quantify it?





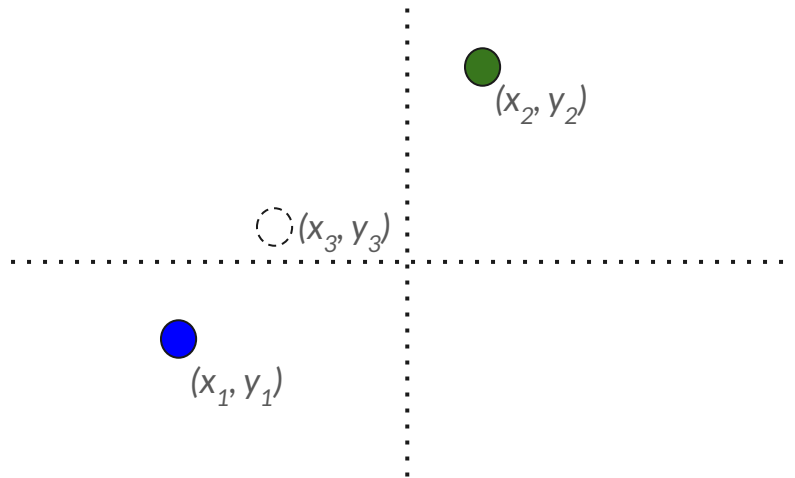
Distance (or Proximity) metric

- Let's we are give two data points: one, the **blue**, and the other is the **green** circle.
- Now lets, we are given a new circle, and we are given the task to label it with either blue or green.
- Visually, it looks to be closer to the blue, dot; but how do we quantify it?
- We can use a proximity or distance metric.



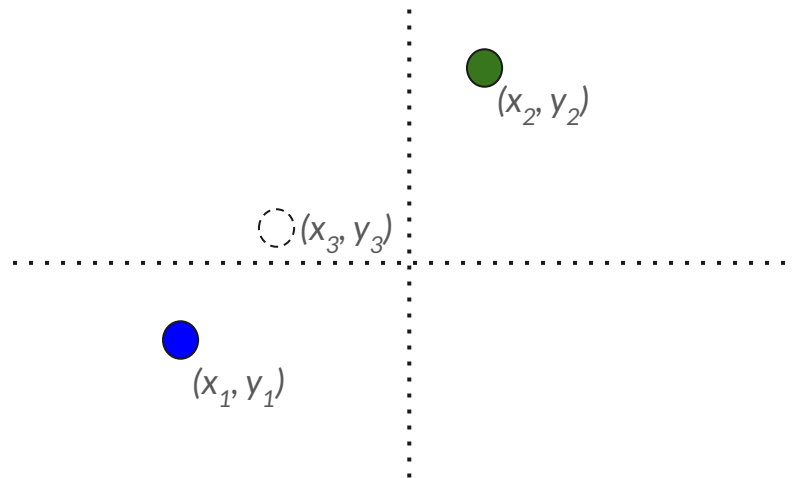
Distance (or Proximity) metric

- We can use a proximity or distance metric.
- These three points are depicted on a 2D plane; right?



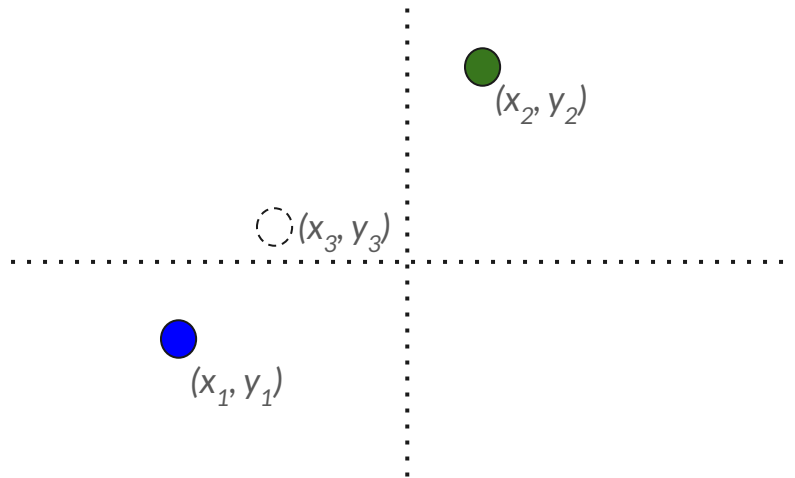
Distance (or Proximity) metric

- We can use a proximity or distance metric.
- These three points are depicted on a 2D plane; right?
- We can use the **Cartesian coordinate system** to quantify the location, and measure their distance; more specifically the Euclidean distance that we learned in our high-school math.



Distance (or Proximity) metric

- We can use a proximity or distance metric.
- These three points are depicted on a 2D plane; right?
- We can use the **Cartesian coordinate system** to quantify the location, and measure their distance; more specifically the Euclidean distance that we learned in our high-school math.
- The Euclidean distance is also known as L2 distance in the DS community

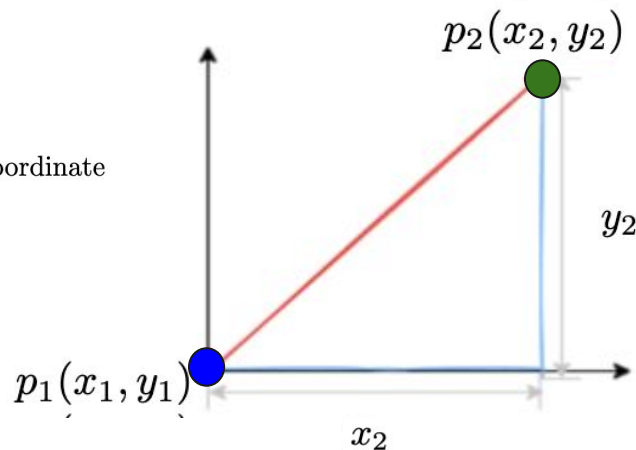


Distance (or Proximity) metric

L2 (or Euclidean) distance: The L2 distance between point $p_1(x_1, y_1)$ and $p_2(x_2, y_2)$ is:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{x_2^2 + y_2^2} \text{ given that } p_1(x_1, y_1) = (0, 0), \text{ the origin of the coordinate}$$

I.e. **L2 distance** is the **diagonal** side of a triangle at the right, also known as **Euclidean distance**

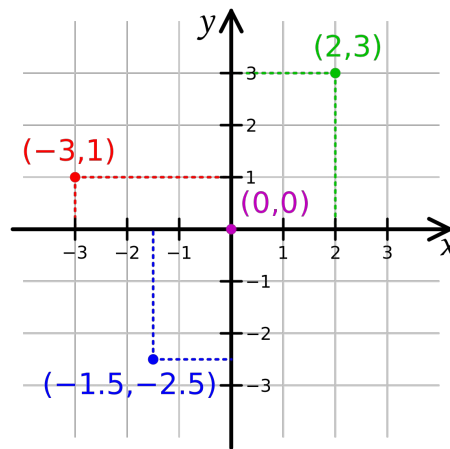


Distance (or Proximity) metric

- L2 (or Euclidean) distance:

Let's
practice

- L2 distance between vectors $[2, 3]$ and $[0, 0]$?
- L2 distance between vectors $[2, 3]$ and $[-3, 1]$?



Distance (or Proximity) metric

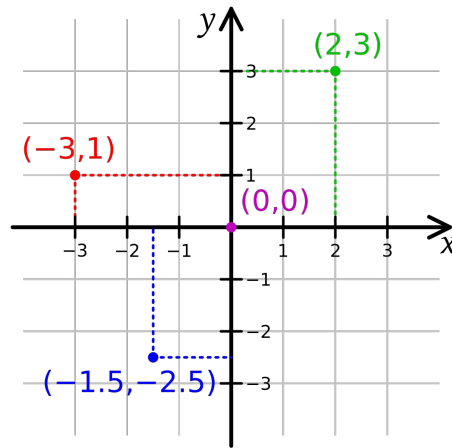
- L2 (or Euclidean) distance:

- L2 distance between vectors $[2, 3]$ and $[0, 0]$ is:

$$\sqrt{(2 - 0)^2 + (3 - 0)^2} = \sqrt{13} = 3.61$$

- L2 distance between vectors $[2, 3]$ and $[-3, 1]$ is:

$$\sqrt{(2 - (-3))^2 + (3 - 1)^2} = \sqrt{29} = 5.39$$





Distance (or Proximity) metric

- We have other distance metrics, such as
- **L1 distance**

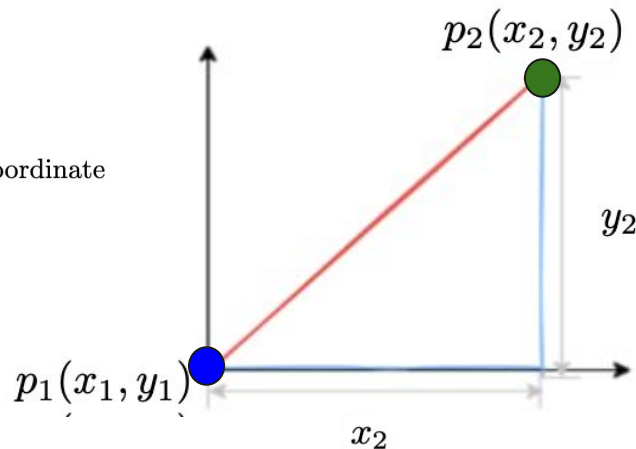
Distance (or Proximity) metric

L1 distance: The L1 distance between point $p_1(x_1, y_1)$ and $p_2(x_2, y_2)$ is:

$$|x_2 - x_1| + |y_2 - y_1|$$

$$= x_2 + y_2 \quad \text{given that } p_1(x_1, y_1) = (0, 0), \text{ the origin of the coordinate}$$

I.e. L1 distance is the summation of the **horizontal** and the **vertical** sides of a triangle at the right.

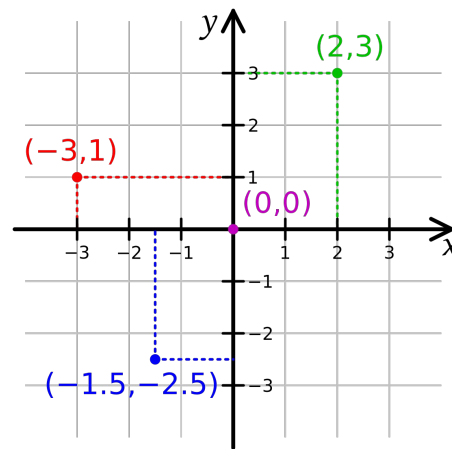


Distance (or Proximity) metric

- L1 distance

Let's
practice

- L1 distance between vectors $[2, 3]$ and $[0, 0]$?
- L1 distance between vectors $[2, 3]$ and $[-3, 1]$?



Distance (or Proximity) metric

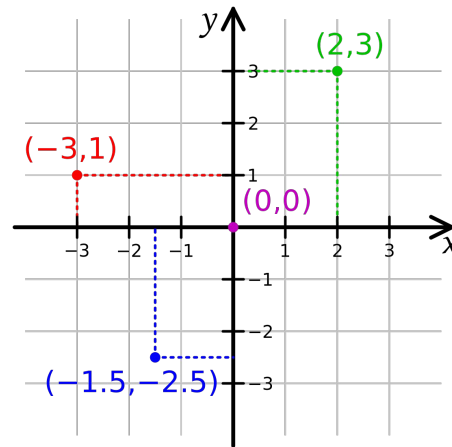
- L1 distance

- L1 distance between vectors $[2, 3]$ and $[0, 0]$ is:

$$|2-0| + |3-0| = 5$$

- L1 distance between vectors $[2, 3]$ and $[-3, 1]$ is:

$$|2 - (-3)| + |3 - 1| = 5 + 2 = 7$$

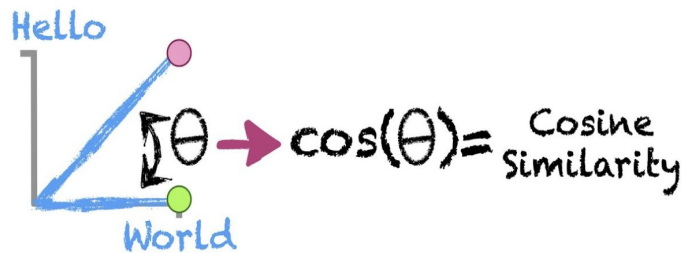




Another unique Distance metric

- We have other distance metrics, such as
- L1 distance, and
- **Cosine distance**

Cosine similarity



Cosine similarity between vectors: $\vec{a}$ & $\vec{b}$ is:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2}$$

$$\|\vec{b}\| = \sqrt{b_1^2 + b_2^2 + b_3^2 + \dots + b_n^2}$$



Cosine distance (angular)

Cosine distance = 1 - Cosine similarity

Cosine distance between vectors: $\vec{a}$ & $\vec{b}$ is:

$$1 - \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

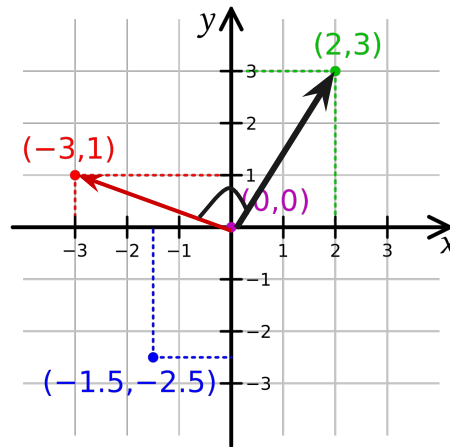
$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2}$$

$$\|\vec{b}\| = \sqrt{b_1^2 + b_2^2 + b_3^2 + \dots + b_n^2}$$

Cosine distance (angular)

Let's
practice

- Cosine similarity between vectors $[2, 3]$ and $[-3, 1]$?
- Cosine distance?

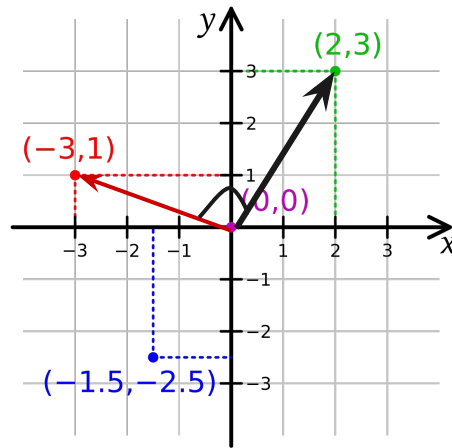


Cosine distance (angular)

- Cosine similarity between vectors $[2, 3]$ and $[-3, 1]$ is :

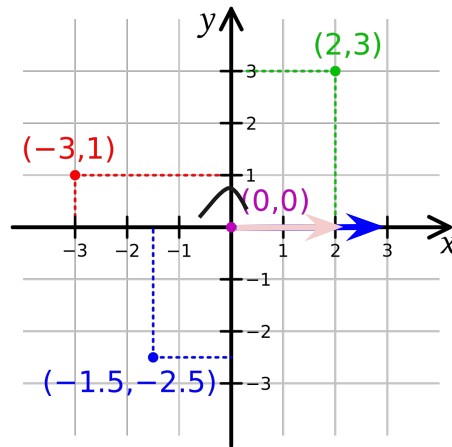
$$\frac{-3}{\sqrt{13}\sqrt{10}} = -0.26$$

- Cosine distance $= 1 - (-0.26) = 1.26$



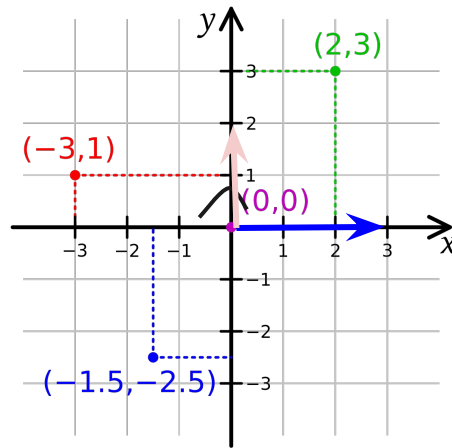
Cosine distance (angular)

- Cosine similarity range : $(-1, +1)$
- Two proportional vectors $[3, 0]$ and $[2, 0]$ (same direction) have a Cosine similarity: 1,
- And Cosine distance = $1 - 1 = 0$



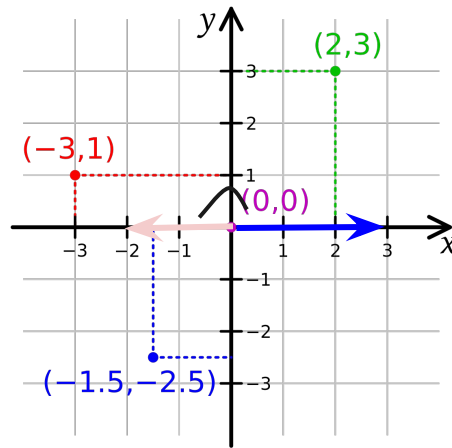
Cosine distance (angular)

- Cosine similarity range : $(-1, +1)$
- Two orthogonal vectors $[3, 0]$ and $[0, 2]$ have a Cosine similarity : 0,
- And Cosine distance = $1 - 0 = 1$



Cosine distance (angular)

- Cosine similarity range : $(-1, +1)$
- Two opposite vectors, $[3, 0]$, $[-2, 0]$, have a Cosine similarity : -1 ,
- And Cosine distance $= 1 - (-1) = 2$
- Cosine distance range is: $(0, 2)$



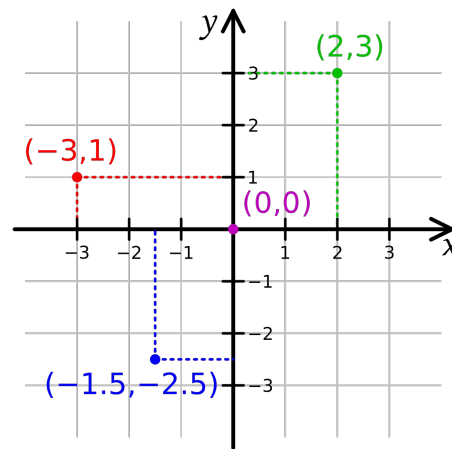


Our first ML Model

- k-NN

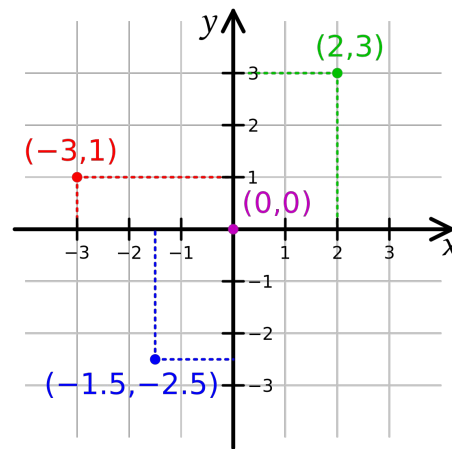
k-NN model

- k-nearest neighbors (k-NN)
 - Supervised learning



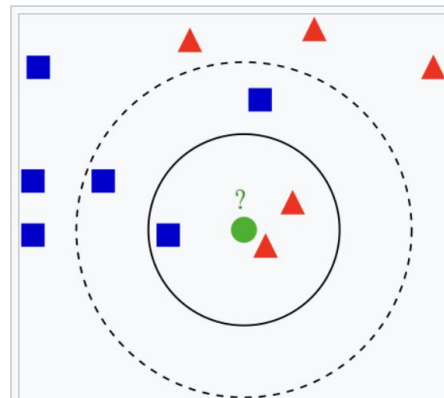
k-NN model

- k-nearest neighbors (k-NN)
 - Supervised learning
 - Non parametric



k-NN model

- k-nearest neighbors (k-NN)
 - Supervised learning
 - Non parametric (distance based method)
 - Both for Classification and Regression solutions



Example of k -NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If $k = 3$ (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If $k = 5$ (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).